BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: 92301

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. The basic cause of ulcers is :

(A) Infection in colon wall located in abdomen
(B) Destruction of the gut wall by hydrochloric acid
(C) Cigarette smoking
(D) Frequent use of spicy food

Q2. The artria in the heart is present:

(A) Below the ventricles
(B) Above the ventricles
(C) On the right side of ventricles
(D) On the left side of ventricles

Q3. The rate of photosynthesis lowers down when :

(A) CO_2 falls down in vascular tissues
(B) O_2 falls down in vascular tissues
(C) CO_2 falls down in mesophyll tissues
(D) O_2 falls down in mesophyll tissues

Q4. Which of the following is a product of calvin cycle?

(A) Fructose
(B) Sucrose
(C) Carbon dioxide
(D) Glucose

Q5. Temporary enzyme-substrate complex is formed when:

(A) Enzyme attaches with products
(B) Enzyme attaches with active site
(C) Enzyme attaches with substrate
(D) Enzyme attaches with prosthetic groups

Q6. During anaphase-II stage the centromere of each chromosome divide due to the pull exerted by :

(A) Equatorial plate
(B) Nuclear envelope
(C) Spindle fibre
(D) Sister chromatids

Q7. In meiosis, synapse and crossing over are main events of:

(A) Prophase-I
(B) Metaphase-I
(C) Metaphase-II
(D) Anaphase-I

Q8. A cell has ribosomes larger in size and nuclear material is surrounded by a bounded nucleus.This cell is an example of:

(A) Bacteria
(B) Cyanobacteria
(C) Prokaryote
(D) Eukaryote

Q9. Xylem cells are tube like and have thick walls and are used for:

(A) Conduction of food and support
(B) Conduction of Water and support
(C) Production of Proteins and support
(D) Production of Fats and support

Q10. The cell wall of yeast is made of:

(A) Chlorophyll
(B) Murein
(C) Chittin
(D) Cellulose

Q11. In scientific method, experiments are conducted to prove or disprove:

(A) Observation
(B) Hypothesis
(C) Biological problem
(D) Conclusion

Q12. The study of classification and naming organisms is called:

(A) Physiology
(B) Taxonomy
(C) Socio-Biology
(D) Morphology

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) How relationship of biology and animal sciences is important?

(ii) Why do five Kindoms can't classify viruses to any kingdom system?

(iii) How Anton van leeuwenhoek observed unicellular creatures? What term he used for them?

(iv) Decribe role of sieve tube in phloem of plants and how companion cells help them to continue their function?

(v) Write importance of mitosis in cell division.

(vi) Differentiate between mitosis and meiosis with the help of following table. Mitosis | Meiosis Metaphase | Metaphase-I Anaphase | Anaphase-I

(vii) What are the key features of lock and key model of enzyme action?

(viii) How ATP is formed in last step of aerobic respiration?

(ix) Column A contain some properties of vitamin C.Write their answer in column B. Column A | Column B Group | Main function | Signs of deficiency | Main sources |

(x) How do stomata contribute to transpiration?

(xi) Explain the role of bicuspid valve in blood circulation.

SECTION "C"

Maximum Marks: 21

> Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [3] How does data help scientists to make hypotheis and conclude?

(b) [3] Musk deer and Indus river dolphin are endangered animal species in Pakistan. Enlist reasons.

QIV.

(a) [4] How ribosome, cell wall, nucleus and plastids differentiate prokaryotic cell and Eukaryotic cell.

(b) [3] Draw diagram of late prophase and indicate position of centromere.

QV.

(a) [3] Differentiate between prosthetic group and coenzymes.

(b) [4] Write steps involved in Krebs cycle.

QVI.

(a) [4] Write any Three risks of chemical fertilizers for soil.

(b) [3] Differentiate between different layers of arteries.